Eya Jabbes

Machine Learning Engineer · Foundation Models for Computational Pathology
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Summary

Machine Learning Engineer with 4 years at InstaDeep, promoted to Tech Lead in 2025. Specializes in computer vision and foundation models for medical imaging and computational pathology. Core engineer on BioNTech's tumor-tissue segmentation model, owning its deployment to production on the client's platform and raising its IoU from 81.6 to 91.4.

Technical Skills

- **Software Engineering:** Python, SQL, Git, CI/CD, pytest, mypy, Claude Code (agentic workflows, custom skills)
- **Deep Learning:** PyTorch, PyTorch Lightning, Hugging Face (Transformers, PEFT), TensorFlow, Scikit-learn, XGBoost
- **Vision Models:** DINOv2, DINOv3, Prov-GigaPath, H-optimus, Hibou, PathDINO, LongViT, Mask2Former, ViT-Adapter
- **Computational Pathology:** H&E whole-slide images, tissue segmentation, MSI prediction, slide-level embeddings
- **Methods:** self-supervised learning, parameter-efficient fine-tuning, domain adaptation, model parity testing
- **MLOps & Deployment:** ONNX Runtime, Docker, MLflow, Neptune, Hydra, FastAPI, GCP, AWS CLI, AIchor (Kubernetes)
- **Languages:** Arabic (native), French (fluent), English (professional)

Professional Experience

InstaDeep, a BioNTech company | *Machine Learning Engineer, promoted to Tech Lead (Sep 2025)* | Sep 2022 – Present

- **H&E Histopathology Image Analysis for BioNTech (Jun 2023 – Present):** Built the project's first gigapixel whole-slide data loader, the data foundation later training pipelines built on, selecting pyramid levels by microns per pixel, and integrated DINOv2 into distributed multi-GPU training.
  - **Segmentation model:** Implemented configurable, swappable ViT foundation-model backbones, proposed and implemented a Mask2Former head (IoU **72 to 81.6**), then fixed a ViT-Adapter fine-tuning bug (IoU **81.6 to 91.4**).
  - **Backbone selection:** Benchmarked 5 frozen ViT foundation models behind one ViT-Adapter and Mask2Former head, comparing pathology-specific (Hibou-B, Hibou-L, PathDINO, H-optimus) with general-purpose DINOv3. Hibou-B (86M parameters) had the best IoU, ahead of the 1.1B H-optimus, and became the project's backbone.
  - **Domain shift and generalization:** Fine-tuned the colorectal model on breast cancer (**88** mean IoU), then trained one joint model that generalizes across both cancer types (**87** mean IoU).
  - **Production deployment:** Owned the PyTorch to ONNX model deployment of the Mask2Former tissue-segmentation model for production inference on the client's platform, with an automated parity check against PyTorch outputs.
  - **Cross-functional collaboration:** Presented results in biweekly meetings with the BioNTech team, ran technical deployment sessions with them, and worked directly with pathologists.
  - **MSI/MSS prediction** (immunotherapy biomarker): Co-developed the first version of a LongViT model predicting tumor microsatellite instability. Against 3 vendors, it ranked first in recall and second in accuracy and F1-score.
  - **Manufacturing success prediction** (sequencing QC): Extracted tile- and slide-level Prov-GigaPath embeddings from 1,195 slides for a LightGBM classifier flagging failed library preps. Tissue-type pooling lowered false positives.
- **Decision Transformer Research, DeepPack (Jan 2023 – Jun 2023):** Co-developed a GPT-2 Decision Transformer for RL-based bin packing on 1.5TB of trajectories from the production SPPO agent. It beat SPPO on 1 of 45 instances, and an XGBoost probe (70.5% vs. 9.5% accuracy on raw vs. embedded states) found the bottleneck in state embeddings.
- **Hiring:** Co-designed ML and deep learning technical tests with the hiring team and interviewed internship candidates.
- **Mentoring:** Mentored a histopathology intern (Feb 2025 – Aug 2025) on foundation-model efficiency research that produced a new knowledge-distillation method.

InstaDeep | *Research Intern* | Feb 2022 – Aug 2022

- **Arabic Lip Reading:** Built a 10-hour multi-dialect Arabic audio-visual dataset (13 speakers) for speech recognition aimed at ALS patients, then fine-tuned the AV-HuBERT model, cutting audio-visual word error rate from 98% to 40%.

Internship (2021): CNN for Alzheimer's staging from MRI (University of Moncton, Canada, remote).

Guest speaker: *AI for Brain Disease* event, National Engineering School of Tunis (ENIT), Dec 2022.

Education

National Engineering School of Tunis (ENIT), Tunis, Tunisia
Master of Research in Artificial Intelligence and Signal Processing, 2021 – 2022

Higher School of Statistics and Information Analysis (ESSAI), Tunis, Tunisia
Engineering Degree in Statistics and Data Analysis, 2019 – 2022